

C++ Programming Internship

Thiranex • Project-Based Internship

P Vedansh | Roll No. 25SCS1003000705 | 3rd Semester • 2CSE32

25 July 2026 - 24 August 2026

👤 ABOUT THE CANDIDATE

Candidate Profile

Personal Details

Name: P Vedansh

Roll Number: 25SCS1003000705

Programme: B.Tech Computer Science & Engineering

Batch: 2025-2029

Internship Details

Semester: 3rd

Section: 2CSE32

Role: C++ Programming Intern

Organisation: Thiranex

Duration: 25 July - 24 August 2026

Thiranex & Project-Based Learning



Practical Focus

Hands-on C++ programming through real-world project development – not just theory, but applied engineering.



Learning Outcome

Converting programming concepts into complete, functional applications – bridging knowledge and execution.



Project-Based Approach

Multiple console applications designed to apply programming fundamentals in structured, meaningful ways.

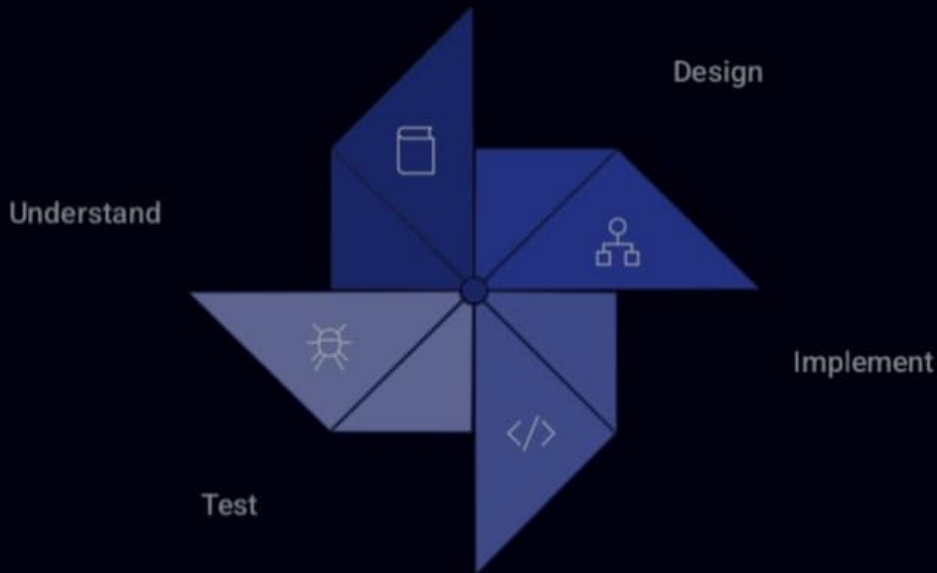


Duration

25 July 2026 – 24 August 2026 · Remote / Project-Based internship format.

From C++ Concepts to Working Applications

Learning C++ syntax and individual programming concepts is different from applying them together to build complete applications. This internship provided hands-on experience in converting programming concepts into structured, functional console-based applications.



Internship Objectives

1

C++ Fundamentals

Strengthen functions, loops, arrays, conditions, and input/output operations.

2

Object-Oriented Programming

Apply classes and objects to model real-world systems effectively.

3

File Handling

Understand and apply file I/O for persistent data storage across sessions.

4

Application Design

Build structured and menu-driven console applications with clean architecture.

5

Problem Solving

Convert requirements into logical and organized C++ solutions.

6

Testing & Debugging

Test program functionality, identify errors, and improve reliability.

Technologies & Tools

C++

Primary language for all project implementations



GCC

GNU Compiler Collection — compile & execute

VS Code

IDE for coding, debugging, and development

GitHub

Version control and project repository management

OOP Concepts

Classes, objects, and encapsulation for structured design

File Handling

Data persistence and file I/O operations

System Architecture



User Input

Menu-driven interface for selections and data entry



Control Flow

Conditions and loops direct execution based on choices



Program Logic

Functions and OOP implement application operations



Data Handling

File handling manages persistent storage and retrieval



Output

Program displays results and feedback to the user

The Project-Based Learning Journey

1 Student Management System

Console application for managing student records – add, update, delete, and display functionality using structured file handling.

2 Bank Management System

Banking application simulating account creation, deposits, withdrawals, and balance inquiries using OOP principles.

3 Library Management System

Library system for managing books, members, issuing and returning books, with integrated search functionality.

4 Tic Tac Toe

Two-player game with 3×3 board, win detection, and replay options – demonstrating logic, arrays, and input validation.

Practical Work & Capabilities Gained

Student Management System

- Structures & file handling
- Menu-driven interface design
- CRUD operations on student records

Bank Management System

- Object-Oriented Programming
- Classes, objects & encapsulation
- Account & transaction handling

Library Management System

- Classes & file handling
- Searching algorithms
- Member and book management

Tic Tac Toe

- Arrays & game logic
- Win detection algorithms
- Input validation & control flow

Successfully Completed

Thiranex C++ Programming Internship

4

Projects Built

Console applications developed end-to-end

1

Month

Intensive project-based learning period

6+

C++ Concepts

Applied across all four projects

P Vedansh · 25 July 2026 – 24 August 2026 · Certificate ID: THX-JUL2526-371

Thank You

P Vedansh

Roll No. 25SCS1003000705

B.Tech CSE · 3rd Semester · 2CSE32

Thiranex C++ Programming Internship

github.com/pvedansh1110/2025-29_P_Vedansh_25SCS1003000705_3rd_2CSE32